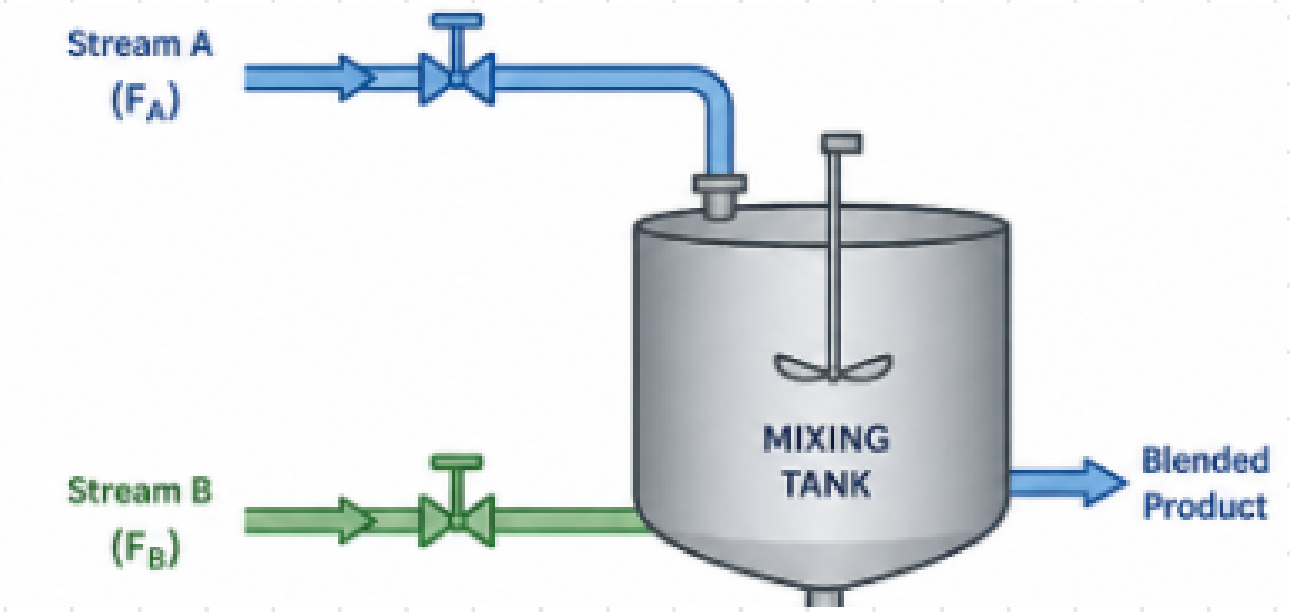
Team Details :

Team GKP-II presents Process Control Project



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Selected Problem Statement :

"Design and simulate a ratio control system for a blending process, where two inlet streams must maintain a fixed composition despite flow disturbances, and compare its performance with independent flow control loops."

Overview of solution :

This work focuses on the design and simulation of **ratio control and independent flow** control for a blending process using MATLAB/Simulink to compare their performance under disturbances.

- **Ratio Control :**

Stream 1 (controlled flow) regulated by a valve, and Stream 2 (wild flow) which is measured. In ratio control, we control one flow based on the other to keep a fixed ratio.

Controlled Flow = Ratio × Wild Flow, with feedforward for disturbance rejection

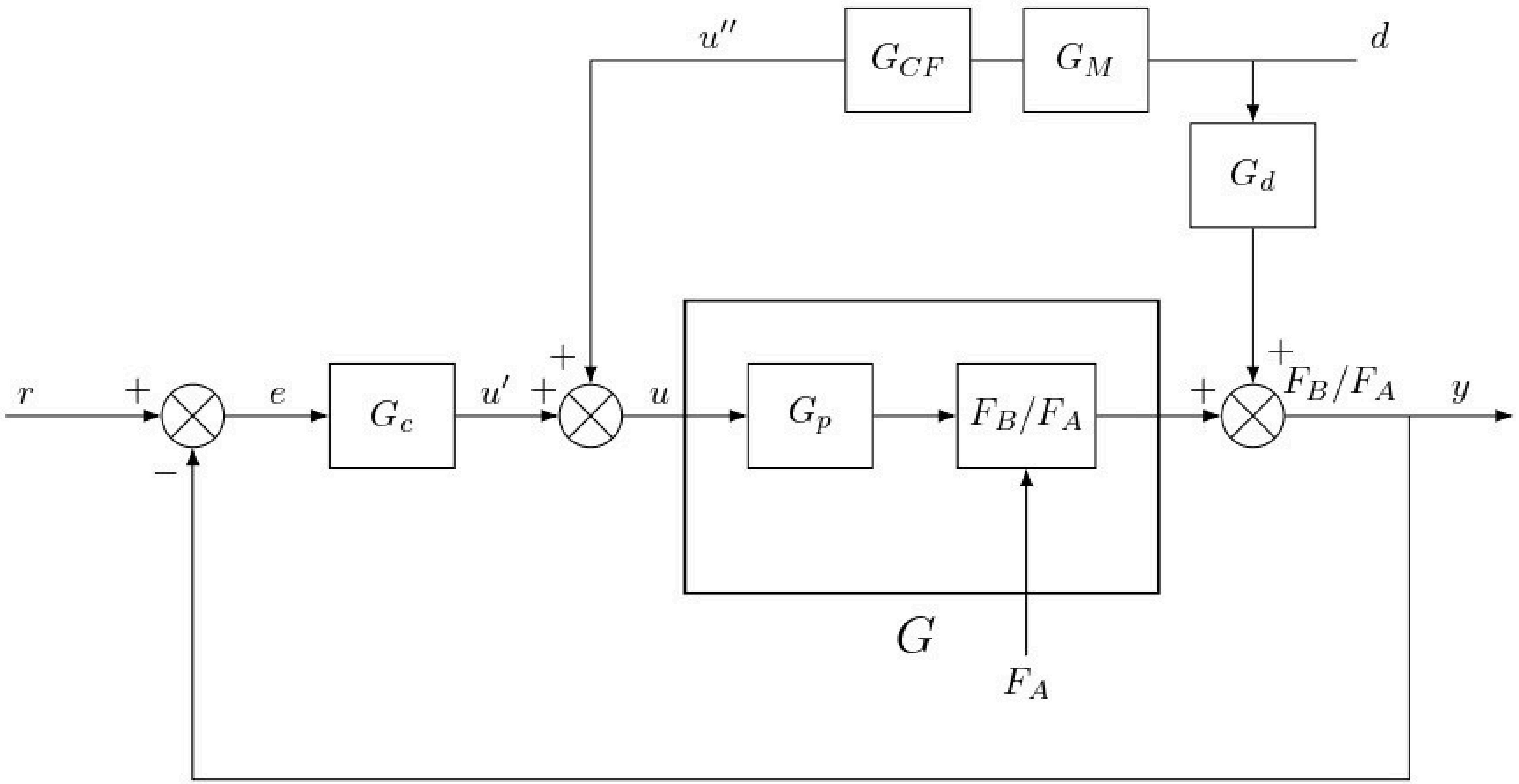
- **Independent Flow Control** :

It is also implemented, where both streams are controlled separately using individual controllers, with the relation **FB = FA × R**. Set-point FA (Stream 1) and target ratio R are defined;

FB = FA × R gives the set-point for Stream 2.

Results show that ratio control provides better composition control and disturbance handling.

Block Diagram and Transfer Functions :



The block diagram : shows the ratio control system with feedback and feedforward control.

System Transfer Functions

$$g_d(s) = \frac{0.001237}{s + 3.6 \times 10^{-4}}$$

$$g_{cf}(s) = \frac{-0.002195}{s + 8.6 \times 10^{-4}}$$

$$g_p(s) = \frac{0.563s + 2.3 \times 10^{-5}}{s + 9.09 \times 10^{-5}}$$

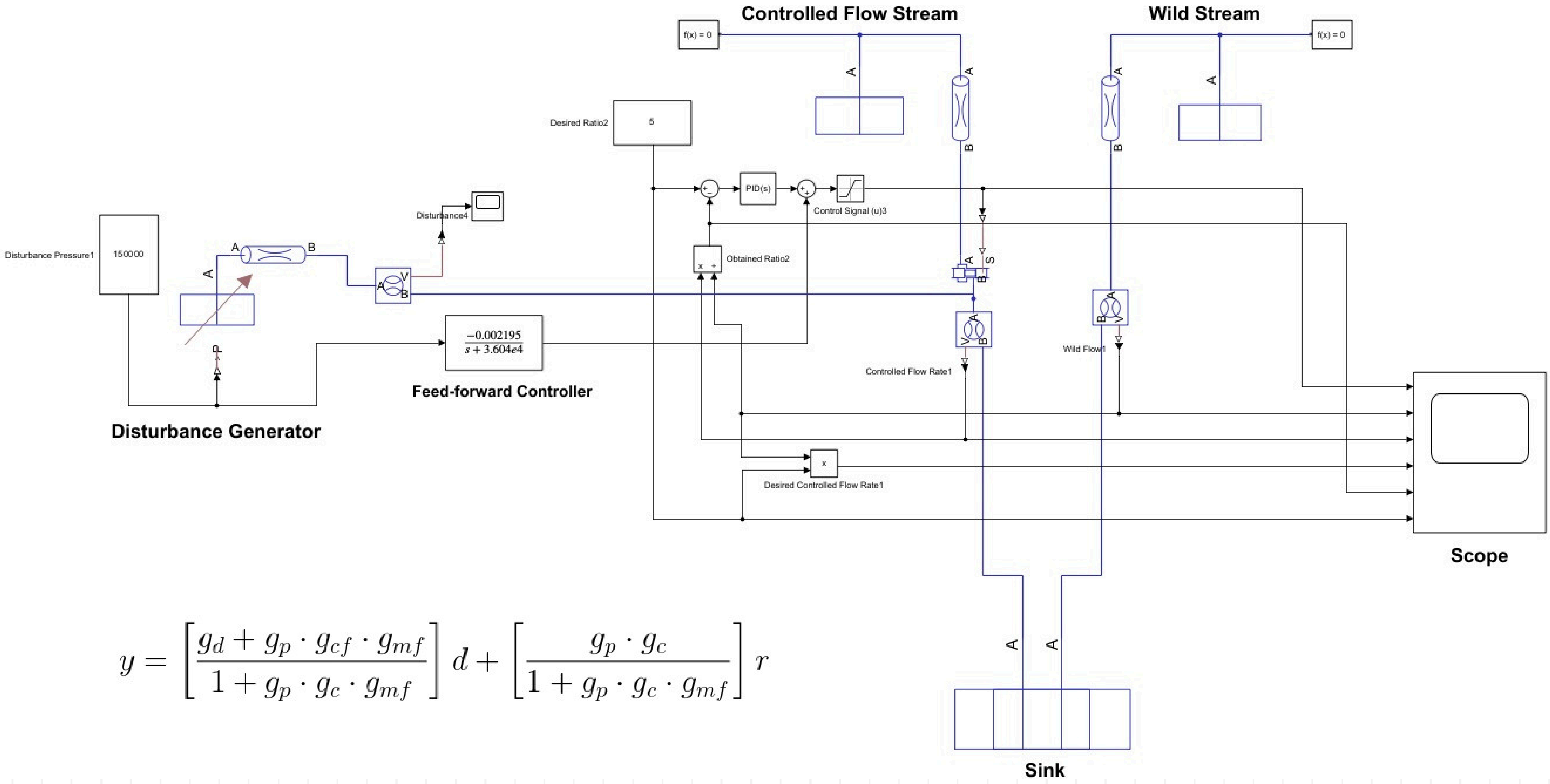
$$g_c(s) = 7.5 + \frac{10}{s} = \frac{7.5s + 10}{s}$$

$$g_{mf}(s) = 1$$

Final System Equation

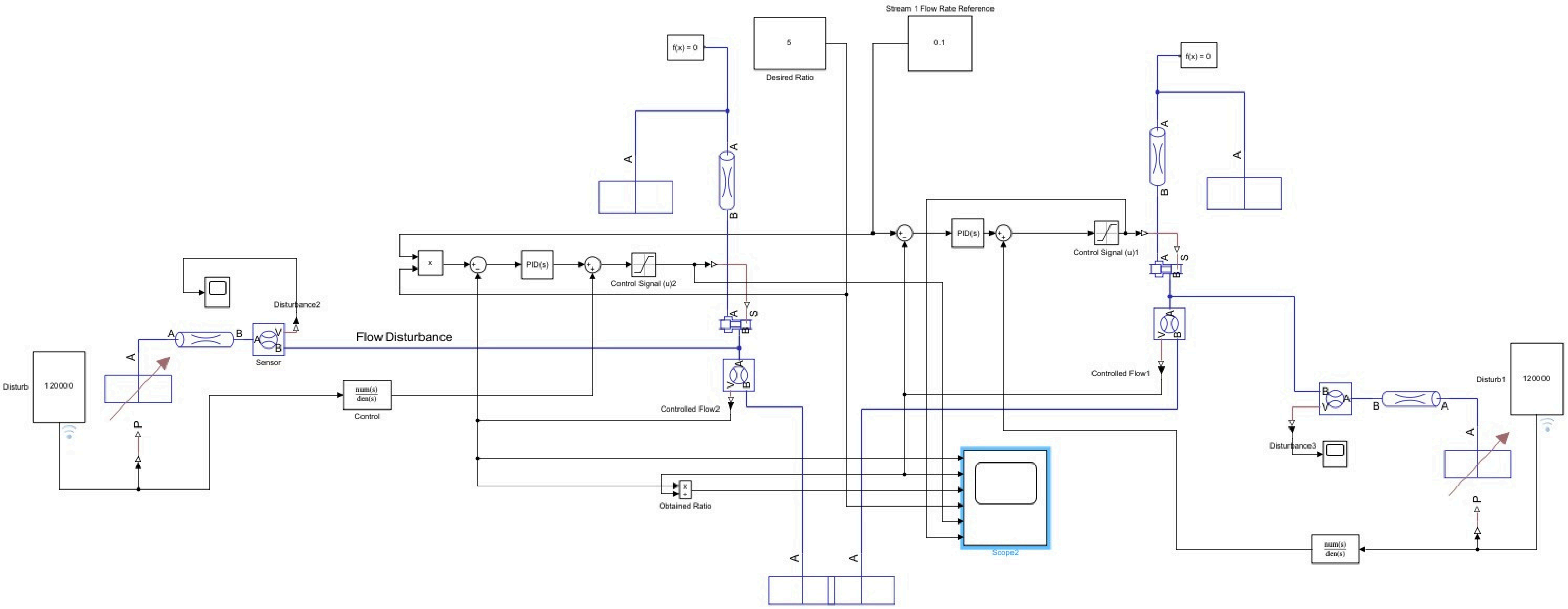
$$y = \left(\frac{1.237 \times 10^{-6}s^2 + 3.42 \times 10^{-7}s + 7.84 \times 10^{-11}}{5.22s^4 + 7.94s^3 + 2.05s^2 + 1.25 \times 10^{-4}s + 7.15 \times 10^{-8}}\right) d + \left(\frac{4.2225s^2 + 5.63017s + 2.3 \times 10^{-4}}{5.2225s^2 + 5.63026s + 2.3 \times 10^{-4}}\right) r$$

Ratio Control Diagram :

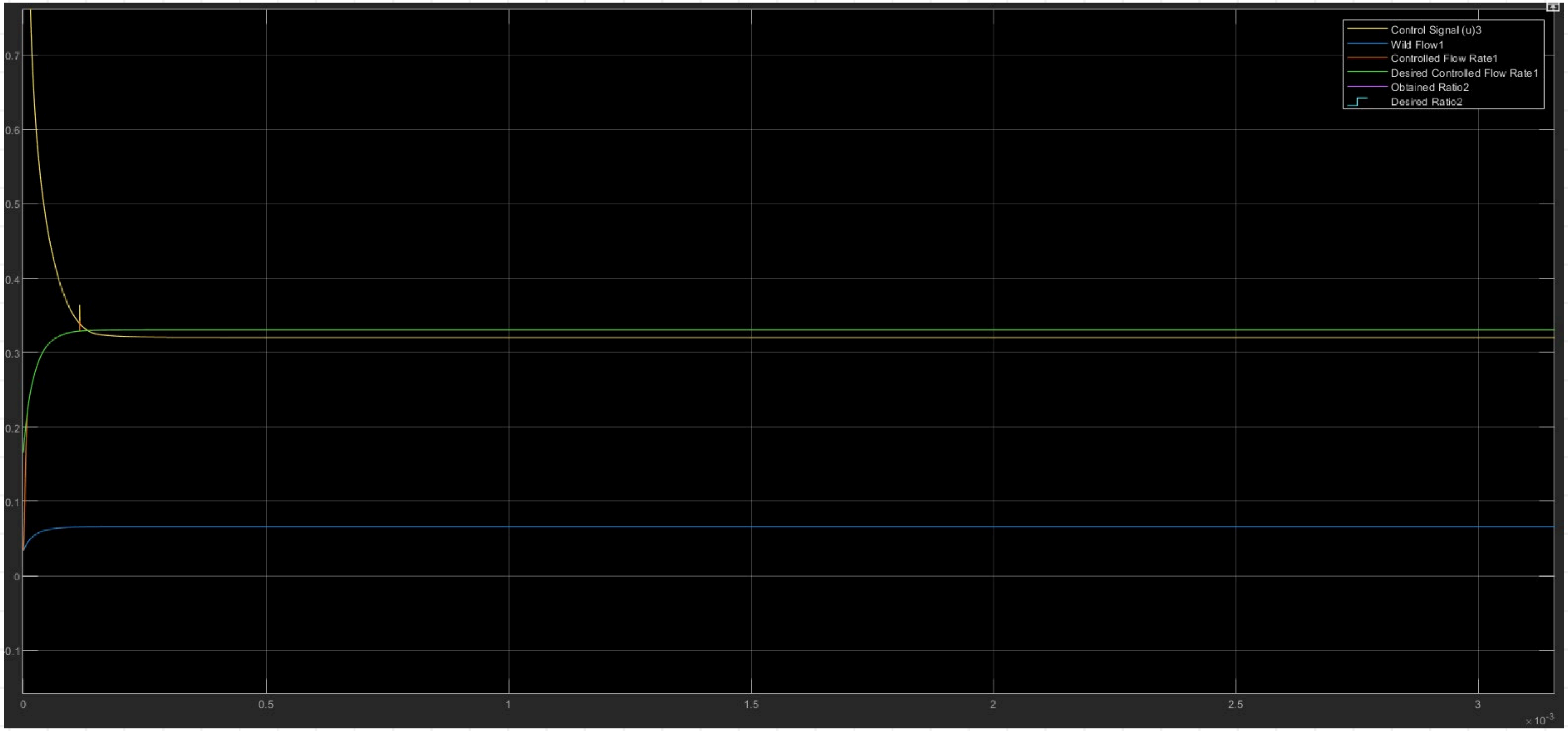


$$y = \left[\frac{g_d + g_p \cdot g_{cf} \cdot g_{mf}}{1 + g_p \cdot g_c \cdot g_{mf}} \right] d + \left[\frac{g_p \cdot g_c}{1 + g_p \cdot g_c \cdot g_{mf}} \right] r$$

Independent Flow Control Diagram :



Ratio Controller’s Output with parameters :



Control signal = 0.321

Wild flow rate-> 66.175 L/s

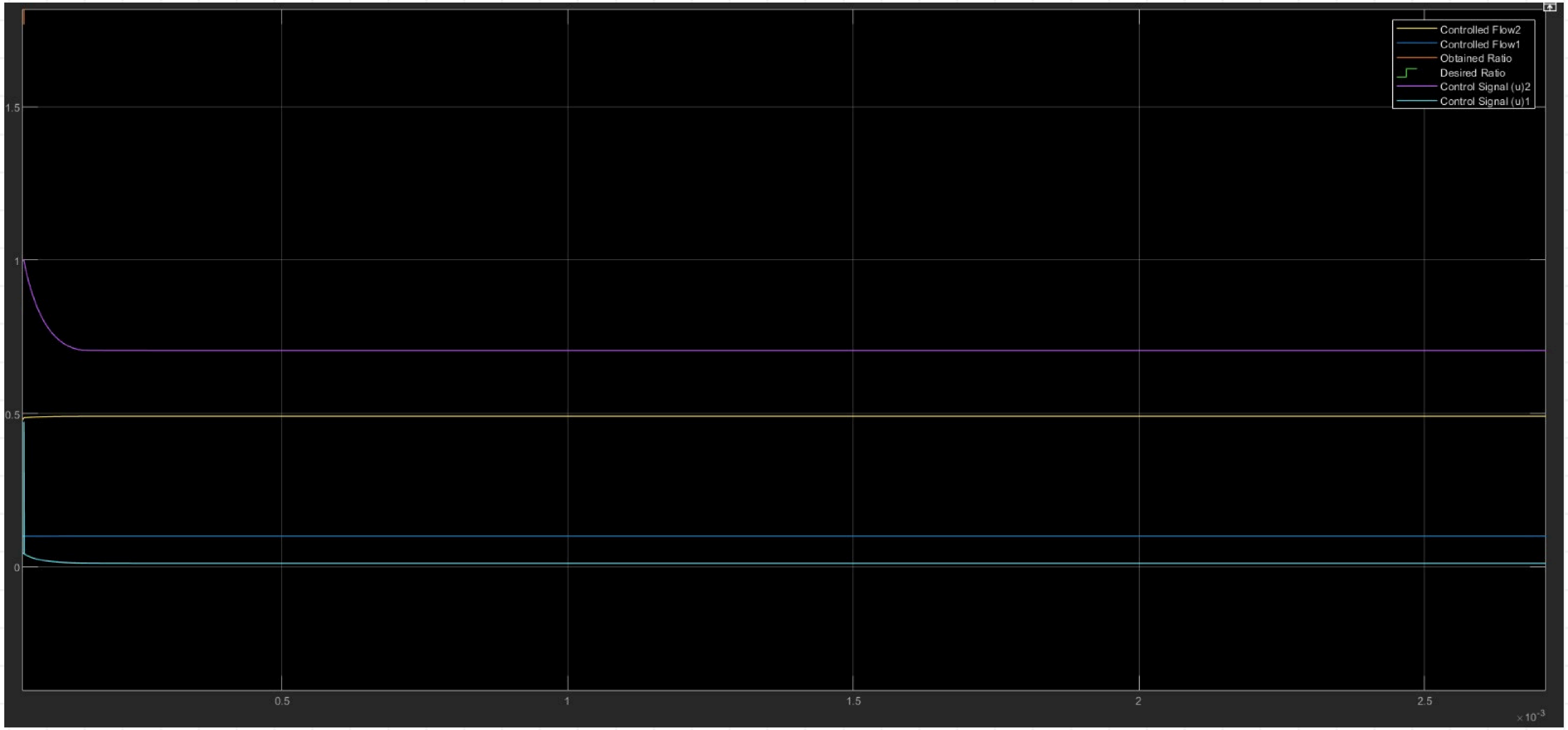
Controlled flow rate ->
330.795 L/s

Desired control flow rate ->
330.9 L/sec

Obtained ratio -> 4.999

Desired ratio -> 5.000

Independent Controller's Output with parameters :



CS 1 -> 0.7050

CS 2 -> 0.0115

Obtained flow rate 1 -> 490.50 L/sec

Obtained flow rate 2 -> 99.29 L/sec

Obtained ratio -> 4.912

Desired ratio-> 5.00

Results and Discussions :

01 RESULTS

RATIO CONTROL	
Wild Flow (F_2 / FB)	66.175 L/s
Controlled Flow (F_1 / FA)	330.795 L/s
Desired Flow	330.9 L/s
Obtained Ratio	4.999
Desired Ratio	5.000

INDEPENDENT FLOW CONTROL	
Flow 1 (F_1 / FA)	490.50 L/s
Flow 2 (F_2 / FB)	99.29 L/s
Obtained Ratio	4.912
Desired Ratio	5.000
Steady-State Error	0.088 (1.76%)

02 DISCUSSION

Accuracy & Steady-State Error

Ratio Control achieved $F_1 = R \times F_2 = 5 \times 66.175 = 330.875$ L/s (error: 0.001, 0.02%). Independent Control produced a ratio of 4.912 — an error of 0.088 (1.76%). Ratio Control is ~88× more accurate owing to feedforward coupling.

Disturbance Rejection

Ratio Control automatically compensates wild-flow (F_2) variations by recalculating $F_1 = R \times F_2$ in real time, maintaining the ratio under disturbances. Independent Control relies on delayed feedback, causing transient ratio deviations.

Dynamic Response & Coordination

Ratio Control inherently synchronises both streams — any F_2 change is immediately reflected in F_1's setpoint, eliminating phase lag. Independent Control cannot coordinate simultaneously, causing transient decoupling and ratio overshoot.

Practical Implications

In industrial blending (chemical reactors, fuel mixing), even 1–2% ratio error compromises quality and safety. Ratio Control reduces instrumentation overhead, simplifies tuning, and is inherently robust for variable wild-stream processes.

Difficulties :

- Disturbance Difficulty:
  Handling disturbances is difficult because sudden changes in flow can cause deviation from the desired output, and effective rejection is limited by actuator (valve) constraints.

- Proper valve selection and tuning is challenging due to nonlinearity and flow range limitations.

- Maintaining the control signal within the 0–1 saturation limit restricts achievable flow rates.

- Limited operating range of controlled flow makes it difficult to handle large variations in ratio and wild flow.

- Disturbance rejection is constrained by actuator (valve) limits.

- Accurate model linearization is required; errors can affect controller performance.

Future Improvements :

1. Add Cascade Control which can handle valve non-Linearities+Fast Disturbances
2. Use Total Flow + Ratio Control (2-DOF structure) to control production rate and composition
3. Make Feedforward Adaptive (Gain Scheduling) to make feedforward stay accurate across the operating range
4. Replace PID with IMC-based tuning. IMC offers better performance in long dead times, high nonlinearity, or when better set-point tracking and disturbance rejection situations
5. Move towards MPC (Model Predictive Control) to handle multiple constraints, interacting flows and is generally best for large systems as it predicts future behaviour